(a) cached batch
ratio outside band
gradient zeroed
fraction of tokens
0.2
0.1
0.0
ε=0.05
ε=0.2
ε=0.5
(b) KL / token
0.000
−0.005
−0.010
βKL=0
βKL=0.1
βKL=0.2
1 2 3 4 5 6 7 8
update
(c) reward-model score
2
0
1 2 3 4 5 6 7 8
update